Zhuwei Qin

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Education

George Mason University — Feb. 2017 – Aug. 2020
Ph.D. in Electrical and Computer Engineering — Fairfax, VA, USA

Oregon State University — Sept. 2014 – Dec. 2016
M.S. in Electrical and Computer Engineering — Corvallis, OR, USA

Tianjin University of Science & Technology — Aug. 2010 – June 2014
B.E. in Automation — Tianjin, China

Research Interests

Efficient AI Computing

- Edge & Mobile AI Model Acceleration
- LLM Compression & Adaptation
- Federated & Decentralized Learning

Intelligent Sensing & Neural-Machine Interfaces

- Neural-Machine Interfaces
- Real-Time Gesture Recognition & Rehabilitation Systems
- Wearable/Edge-Based Biomedical Sensing

AI for Education

- Human-centered AI Systems
- Intelligent Tutoring Platforms for Engineering Education
- Teaching Support Platforms for Neurodiverse Students

Experience

San Francisco State University — Aug. 2026 – Present
Associate Professor, Computer Engineering, College of Science & Engineering — San Francisco, CA

San Francisco State University — Aug. 2020 – Aug. 2026
Assistant Professor, Computer Engineering, College of Science & Engineering — San Francisco, CA

Publications

Summary: 38 published/accepted publications, including 26 since joining SFSU (2020–Present), with 5+ manuscripts under review. Total citations: 900+; h-index: 12; 100+ citations per year on Google Scholar.

Conference Papers

[1] Songlin Zhao, Michael Pitts, and **Zhuwei Qin**. "EfficientXpert: Efficient Domain Adaptation for Large Language Models via Propagation-Aware Pruning". *in Proceedings of the Fortieth Conference on Neural Information Processing Systems, Workshop on On-Device Intelligence*. Australia, Dec. 2026.

[2] Peter Chudinov, Zhenyu Lin, Jay Motamarry, Srihita Panati, Xiaorong Zhang, and **Zhuwei Qin**. "NeuroEdge: Real-Time Hand Gesture Recognition with High-Density EMG Using Deep Learning at the Edge". *in Proceedings of the IEEE Biomedical Circuits and Systems Conference (BioCAS)*. Korea, Oct. 2026.

[3] Zhaoshuo Jiang, **Zhuwei Qin**, and Tao He. “Interdisciplinary Research and Education through the Development of AI-assisted Home Health Monitoring Systems”. *in Proceedings of the ASEE Annual Conference & Exposition*. Charlotte, NC, June 2026.

[4] Yiyi Wang, Mary Requa, **Zhuwei Qin**, and Chanel-Taylor Guevarra Medeiros. “Concept-mapping the struggles and strengths of neurodiverse students in engineering: a comparison between expert- and AI-generated trees”. *in Proceedings of the ASEE Annual Conference & Exposition*. Charlotte, NC, June 2026.

[5] Yiyi Wang, Mary Requa, **Zhuwei Qin**, and Adrian Arnaud. “Qualitative Analysis of Neurodivergent Students’ Experience in Engineering”. *in Proceedings of the ASEE Annual Conference & Exposition*. Charlotte, NC, June 2026.

[6] **Zhuwei Qin**, Jennifer Trainor, Zhaoshuo Jiang, and Jose Torres. “Meeting Students’ Needs for Responsible AI: Engineering Student Perspectives from an HSI Public Institution”. *in Proceedings of the ASEE Annual Conference & Exposition*. Charlotte, NC, June 2026.

[7] **Zhuwei Qin**, Zhaoshuo Jiang, Jennifer Trainor, Andre Bouvier, and Jose Torres. “Introducing the AI Collaborative (AICO) Platform: A Pilot Implementation in Mechanics of Solids”. *in Proceedings of the ASEE Annual Conference & Exposition*. Charlotte, NC, June 2026.

[8] Jiahua Tang, Philip Liang, and **Zhuwei Qin**. “EdgeEMG: On-Device Neural Network Training for Real-Time EMG Pattern Recognition”. *in Proceedings of IEEE-EMBS International Conference on Body Sensor Networks*. Nov. 2025, pp. 1–4.

[9] Lauren Gan, Oscar Vazquez, Nicholas Dalton, **Zhuwei Qin**, Xiaorong Zhang, and David Quintero. “Development of a Soft, Spring-like Finger Tendon Exo-Glove Design with a Four-Output Cable-Driven Linear Servomechanism”. *in Proceedings of International Conference on Rehabilitation Robotics*. May 2025, pp. 1738–1743.

[10] Sicheng Liu, Qun Wang, **Zhuwei Qin**, Weishan Zhang, Jingyi Wang, and Xiang Ma. “IRS Assisted Decentralized Learning for Wideband Spectrum Sensing”. *in Proceedings of International Symposium on Intelligent Computing and Networking*. Mar. 2025, pp. 507–523.

[11] Ankita Mukherjee, Helena Alvarez, Xiaorong Zhang, **Zhuwei Qin**, and Charmayne Mary Lee Hughes. “DRome: A Deep Learning-based Mobile Vision System for Real-time Range of Motion Evaluation”. *in Proceedings of IEEE RAS EMBS International Conference on Biomedical Robotics and Biomechatronics*. Heidelberg, Germany, Sept. 2024, pp. 315–320.

[12] Junyu Lu, Amir Modan, Zhenyu Lin, Charmayne Mary Lee Hughes, Xiaorong Zhang, and **Zhuwei Qin**. “mMyoHMI: Real-time EMG-based Pattern Recognition with On-device Learning Adaptation”. *in Proceedings of IEEE RAS EMBS International Conference on Biomedical Robotics and Biomechatronics*. Heidelberg, Germany, Sept. 2024, pp. 747–752.

[13] **Zhuwei Qin**, Xiaorong Zhang, David Quintero, Wenshen Pong, Yiyi Wang, Jenna Wong, and Robert Petrulis. “Engaging Community College Students in Artificial Intelligence Research through an NSF-Funded Summer Research Internship Program”. *in Proceedings of the ASEE Annual Conference & Exposition*. Portland, OR, June 2024.

[14] Kang Jun Bai, Hao Jiang, **Zhuwei Qin**, and Clare Thiem. “Temporal-encoded 6T-RRAM with Bidirectional Control for Future Neuromorphic Systems”. *in Proceedings of the International Symposium on Quality Electronic Design*. San Francisco, CA, Apr. 2024, pp. 1–6.

[15] Xiaorong Zhang, David Quintero, Fatemeh Khalkhal, Zhaoshuo Jiang, **Zhuwei Qin**, Jenna Wong, Yiyi Wang, Wenshen Pong, and Robert Petrulis. “Development and First-Year Outcomes of a NSF-Funded Summer Research Internship Program to Engage Community College Students in Engineering Research”. *in Proceedings of the ASEE Annual Conference & Exposition*. Baltimore, MD, June 2023.

[16] David Quintero, Xiaorong Zhang, **Zhuwei Qin**, et al. “A Collaborative Multidisciplinary Engineering Team Experience with Hierarchical Mentoring from High School through Undergraduate Students within a Summer Research Internship Program”. *in Proceedings of the American Society for Engineering Education 2023 Pacific Southwest Conference*. Los Angeles, CA, Apr. 2023.

[17] Zhenyu Lin, Philip Liang, Xiaorong Zhang, and **Zhuwei Qin**. “Toward Robust High-Density EMG Pattern Recognition using Generative Adversarial Network and Convolutional Neural Network”. *in Proceedings of the IEEE International Conference on Neural Engineering*. May 2023, pp. 1–5.

[18] Junyu Lu, Philip Liang, Jin Chul Rhim, Xiaorong Zhang, and **Zhuwei Qin**. “EffiE: Efficient Convolutional Neural Network for Real-Time EMG Pattern Recognition System on Edge Devices”. *in Proceedings of the IEEE International Conference on Neural Engineering*. May 2023, pp. 1–5.

[19] Weishan Zhang, Yue Wang, Fuxun Yu, **Zhuwei Qin**, Xiang Chen, and Zhi Tian. “Wideband Spectrum Sensing based on Collaborative Multi-Task Learning”. *in Proceedings of the IEEE International Conference on Communications Workshops*. May 2022, pp. 1–6.

[20] Fuxun Yu, Weishan Zhang, **Zhuwei Qin**, Zirui Xu, Di Wang, Chenchen Liu, Zhi Tian, and Xiang Chen. “Fed2: Feature-Aligned Federated Learning”. *in Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. Aug. 2021, pp. 2066–2074.

[21] **Zhuwei Qin** and Hao Jiang. “Towards Context-aware Distributed Learning for CNN in Mobile Applications”. *in Proceedings of IEEE/ACM Symposium on Edge Computing*. Nov. 2020.

[22] Xiang Chen and **Zhuwei Qin**. “Exploring Decentralized Collaboration in Heterogeneous Edge Training”. *in Proceedings of IEEE/ACM Symposium on Edge Computing*. Nov. 2020, pp. 450–453.

[23] Chenchen Liu, Fuxun Yu, **Zhuwei Qin**, and Xiang Chen. “Enabling Efficient ReRAM-based Neural Network Computing via Crossbar Structure Adaptive Optimization”. *in Proceedings of ACM/IEEE International Symposium on Low Power Electronics and Design*. Aug. 2020, pp. 1–6.

[24] Fuxun Yu, **Zhuwei Qin**, Di Wang, Ping Xu, Chenchen Liu, Tian Zhi, and Xiang Chen. “DCCNN: Computational Flow Redefinition for Efficient CNN Inference through Model Structural Decoupling”. *in Proceedings of the Design, Automation and Test in Europe Conference*. Mar. 2020, pp. 1097–1102.

[25] **Zhuwei Qin**, Fuxun Yu, and Xiang Chen. “Task-Adaptive Incremental Learning for Intelligent Edge Devices”. *in Proceedings of the IEEE/ACM Symposium on Edge Computing*. Nov. 2019, pp. 340–341.

[26] Fuxun Yu, **Zhuwei Qin**, Chenchen Liu, and Xiang Chen. “Interpreting and Evaluating Neural Network Robustness”. *in Proceedings of the International Joint Conference on Artificial Intelligence*. Aug. 2019, pp. 4199–4205.

[27] **Zhuwei Qin**, Fuxun Yu, Chenchen Liu, and Xiang Chen. “Functionality-Oriented Convolutional Filter Pruning”. *in Proceedings of the 30th British Machine Vision Conference*. Sept. 2019.

[28] **Zhuwei Qin**, Fuxun Yu, Chenchen Liu, and Xiang Chen. “CAPTOR: A Class Adaptive Filter Pruning Framework for Convolutional Neural Networks in Mobile Applications”. *in Proceedings of the Asia and South Pacific Design Automation Conference*. Jan. 2019, pp. 444–449.

[29] **Zhuwei Qin**, Fuxun Yu, Chenchen Liu, and Xiang Chen. “Demystifying Neural Network Filter Pruning”. *in Proceedings of the Conference on Neural Information Processing Systems, Workshop on Compact Deep Neural Network Representation with Industrial Applications (NeurIPS Workshop)*. Dec. 2018.

[30] Fuxun Yu, **Zhuwei Qin**, Chenchen Liu, and Xiang Chen. “Distilling Critical Paths in Convolutional Neural Networks”. *in Proceedings of the Conference on Neural Information Processing Systems, Workshop on Compact Deep Neural Network Representation with Industrial Applications (NeurIPS Workshop)*. Dec. 2018.

[31] Zirui Xu, **Zhuwei Qin**, Fuxun Yu, Chenchen Liu, and Xiang Chen. “DiReCt: Resource-Aware Dynamic Model Reconfiguration for Convolutional Neural Network in Mobile Systems”. *in Proceedings of the International Symposium on Low Power Electronics and Design (ISLPED)*. July 2018, pp. 1–6.

[32] **Zhuwei Qin**, Zirui Xu, Qide Dong, Yiran Chen, and Xiang Chen. “VoCaM: Visualization Oriented Convolutional Neural Network Acceleration on Mobile System”. *in Proceedings of the International Conference on Computer-Aided Design*. Nov. 2017, pp. 835–840.

[33] Jiachen Mao, **Zhuwei Qin**, Zirui Xu, Kent Nixon, Xiang Chen, Hai Li, and Yiran Chen. “AdaLearner: An Adaptive Distributed Mobile Learning System for Neural Networks”. *in Proceedings of the International Conference on Computer-Aided Design*. Nov. 2017, pp. 291–296.

Journal Articles

[34] Fuxun Yu, Zirui Xu, **Zhuwei Qin**, and Xiang Chen. “Privacy-Preserving Federated Learning for Transportation Mode Prediction based on Personal Mobility Data”. *High-Confidence Computing* (Sept. 2022).

[35] **Zhuwei Qin**, Fuxun Yu, Zirui Xu, Chenchen Liu, and Xiang Chen. “CAPTORX: A Class-Adaptive Convolutional Neural Network Reconfiguration Framework”. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (Mar. 2022), pp. 530–543.

[36] Fuxun Yu, **Zhuwei Qin**, Chenchen Liu, Di Wang, and Xiang Chen. “REIN the RobuTS: Robust DNN-based Image Recognition in Autonomous Driving Systems”. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (June 2021), pp. 1258–1271.

[37] Zirui Xu, Fuxun Yu, **Zhuwei Qin**, Chenchen Liu, and Xiang Chen. “DiReCtX: Dynamic Resource-Aware CNN Reconfiguration Framework for Real-Time Mobile Application”. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* (Feb. 2021), pp. 246–259.

[38] **Zhuwei Qin**, Fuxun Yu, Chenchen Liu, and Xiang Chen. “How convolutional neural network see the world - A survey of convolutional neural network visualization methods”. *Mathematical Foundations of Computing* 1.2 (May 2018), pp. 149–180.

Funding

Summary: 13 funded/completed projects totaling $2.0M overall: 5 as PI (about $221K) and 8 as Co-PI (about $1.81M).

Completed

[1] Sony Sensing Solution University Program
Project title: Efficient and Robust Deep Learning Models for Neural-Machine Interfaces with Spresense Microcontroller.
Role: PI
Project Period: 09/2024–08/2025
Approved budget: $14,118; Hardware Donation: $6,696

[2] Sony Sensing Solution University Program
Project title: Neural2: Efficient Deep Neural Network for Neural-controlled Bionic Arm with Spresense Microcontroller.
Role: PI
Project Period: 09/2023–09/2024
Approved budget: $26,600

[3] SF State Research, Scholarly and Creative Activity (RSCA) Fund 2023
Project title: Intelligent Wireless Spectrum Monitoring with Decentralized On-device Learning
Role: PI
Project Period: 2024–2025
Approved budget: $18,500

[4] SF State Research, Scholarly and Creative Activity (RSCA) Fund 2022
Project title: Towards Collaborative On-device Intelligence by Interpreting Black Box of Deep Neural Networks
Role: PI
Project Period: 2023–2024
Approved budget: $11,915

[5] National Science Foundation
Project title: ELITE: Enhancing Learning through Innovative Teaching Equipment
Role: Co-PI (PI: Dr. Zhaoshuo Jiang)
Project Period: 10/2024–09/2025
Approved budget: $196,135

[6] Department of Defense
Project title: Developing new research and teaching capabilities in Understanding Dynamic Systems and Control Theory at SFSU
Role: Co-PI (PI: Dr. Zhaoshuo Jiang)
Project Period: 08/2024–08/2025
Approved budget: $541,514

[7] Department of Defense
Project title: Acquisition of Instrumental Treadmill, Markerless Motion Capture, and Human-Machine Equipment for a Human-Machine Performance Laboratory to Enhance Research in Engineering and Kinesiology at SFSU
Role: Co-PI (PI: Dr. David Quintero)
Project Period: 06/2024–05/2025
Approved budget: $409,434

[8] Ken Fong Translational Research Fund
Project title: Continuous Grasping Adaptation from Multi-Mode Sensory Fusion for a Hand Rehabilitation Exoskeleton
Role: Co-PI (PI: Dr. David Quintero)
Project Period: 2024–2025
Approved budget: $24,664

[9] STEM-NET Faculty Interdisciplinary Collaborative Research SEED Grant Program
Project title: Development of a physical frailty assessment system through a socio-technical lens
Role: Co-PI (PI: Dr. Charmayne Hughes)
Project Period: 2023–2024
Approved budget: $25,000

[10] CSUPERB Industry Partnership Initiative Grant Program
Project title: mAtas: Computer-vision-based System for Post-stroke Range of Motion Evaluation.
Role: Co-PI (PI: Dr. Charmayne Hughes)
Project Period: 2022–2023
Approved budget: $15,000

Funded (In Progress)

[1] California Education Learning Lab
Project title: SCAPE: Student-centered Collaborative AI Platform for Personalized Engineering Learning Experience
Role: PI
Project Period: 03/2025–01/2027
Approved budget: $150,000

[2] National Science Foundation
Project title: Interdisciplinary Research and Education through the Development of AI-assisted Home Health Monitoring Systems
Role: Co-PI (PI: Dr. Zhaoshuo Jiang)
Project Period: 05/2025–07/2027
Approved budget: $199,930

[3] National Science Foundation
Project title: Supporting Neurodiverse Students in Engineering Teamwork through an AI-Driven Platform
Role: Co-PI (PI: Dr. Yiyi Wang)
Project Period: 10/2024–09/2027
Approved budget: $399,979

Submitted (Not Funded)

[1] National Science Foundation
Project title: Advancing Trustworthy AI Systems in Personalized Engineering Education with Transparent Reasoning
Role: PI
Project Period: 06/2026–05/2029
Requested budget: $899,999

[2] CSU's Artificial Intelligence Educational Innovations Challenge
Project title: From Passive Use to Critical Engagement: Integrating GenAI in Engineering Curricula
Role: PI
Project Period: 2025–2026
Requested budget: $59,152

[3] Sony Faculty Innovation Award
Project title: EuRECA: Empower Robust Edge Computing for Neural-Machine Interface by Generative AI
Role: PI
Project Period: 2024–2025
Requested budget: $87,893

[4] NSF Computer and Information Science and Engineering Research Initiation Initiative
Project title: CRII: CNS: RUI: Interpretable Deep Learning for Efficient Federated Learning
Role: Sole PI
Project Period: 2023–2025
Requested budget: $173,048

[5] NSF Convergence Accelerator Phases 1 and 2 for the 2022 Cohort
Project title: NSF Convergence Accelerator Track H: A User-centered Design Approach in the Development of a Technology-enabled Tele-rehabilitation System Designed for Use in Rural communities
Role: Co-PI (PI: Dr. Charmayne Hughes)
Project Period: 2023–2024
Requested budget: $750,000

[6] SF State Research, Scholarly and Creative Activity (RSCA) Fund 2024
Project title: Facilitating Personalized Learning with Real-time Support in STEM Using Large Language Models
Role: PI
Project Period: 2024–2025
Requested budget: $22,750

Teaching

San Francisco State University

- ENGR 859 On-device Machine Learning — *Sp'23, Sp'24, Sp'26*
- ENGR 844 Embedded Systems — *Fa'20*
- ENGR 456 Computer Systems — *Sp'21, Fa'21, Sp'22, Fa'22, Sp'24, Sp'25*
- ENGR 413 AI in Engineering — *Fa'24*
- ENGR 221 Data Structures and Algorithms in Python — *Sp'26*
- ENGR 212 Introduction to Unix and Linux for Engineers — *Sp'24, Sp'26*
- ENGR 213/214 Introduction to C Programming for Engineers/Laboratory — *Fa'20, Sp'21, Fa'21, Sp'22, Fa'22, Sp'23, Fa'23, Sp'25*

George Mason University

- ECE 445 Computer Organization — *Lab Instructor: Sp'20*
- ECE 301 Digital Electronics — *Lab Instructor: Sp'19, Fa'19*

Professional Activities

- **NSF Panelist**
- **Program Committee Member:** Design Automation Conference (DAC 2025, 2026), Asia and South Pacific Design Automation Conference (ASPDAC 2022, 2023)
- **Technical Session Chair:** ACM Great Lakes Symposium on VLSI (GLSVLSI 2021)
- **Organizing Committee Member:** Web Chair, NSF Workshop on Redefining the Future of Computer Architecture from First Principles (2022); Online Organization Chair, NSF Workshop on Processing-In-Memory Technology (2021)
- **Manuscript Reviewer:** AAAI, ASPDAC, IEEE TCASAI, IEEE Cloud Summit, IEEE SOCC, IEEE TSMC, IEEE TEDAES, IEEE TMC, IEEE TCAS-I, IEEE TNNLS, IEEE TCAD, IEEE TCCN, IEEE TC, ACM TECS, Neurocomputing, Journal of Cloud Computing, Neural Computing and Applications
- **SFSU Student Organization Advisor:** Institute of Electrical and Electronics Engineers (IEEE), SFSU Chapter (2023–present), California Solar Regatta SFSU Team (2023–present)
- **SFSU Faculty Judge:** SFSU College of Science and Engineering Showcase (2022, 2023, 2025, 2026), CSU Student Research Competition, Engineering and Computer Science Area (2023, 2024, 2026), SFSU Junior Science and Humanities Symposium (2023, 2025)

Awards

- First Place, The KumoRFM Hackathon Challenge (as faculty advisor) *2025*
- SFSU Campus-wide Winner, Area of Education (Undergrad-Level), CSU Research Competition (as faculty advisor) *2025*
- Honorable Mention Award, Area of Education (Undergrad-Level), CSU Research Competition (as faculty advisor) *2025*
- Third Place Award, Psychology, Human Geography, and Science Education Category, CoSE Project Showcase, SFSU (as faculty advisor) *2025*
- Grand Prize, Sony's Spresense Developer Challenge (as faculty advisor) *2022*
- First Prize, IEEE Services Hackathon *2020*
- Summer 2020 Research Fellowship, GMU, Fairfax, VA *2020*
- Best Poster Award, The 4th ACM/IEEE Symposium on Edge Computing, USA *2019*
- Richard Newton Young Fellow Award, The 54th ACM/IEEE Design Automation Conference, USA *2017*
- OSU Continued Success Scholarship, OSU, Corvallis, OR *2015*
- Thesis Design Award as Outstanding Graduate (20/535), TUST, Tianjin, China *2014*
- Honorable Mention, "TI Cup" Tianjin Electronic Design Competition for College Students, TUST, China *2012*
- Outstanding Individual (25/535), TUST, China *2012*